

Exercise 1: SQL FUNDAMENTALS (Select & Filtering)

1. SELECT *

FROM employees;

Expected Outcome

id	first-name	last-name	department	Salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-08-22	Johannesburg
3	Cathy	Zulu	Finance	65000	2018-07-18	Cape Town
4	David	Mokoena	Marketing	50000	2021-11-05	Durban
5	Eva	Naidoo	IT	72000	2017-09-30	Pretoria

2. Find all unique departments

SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending
SELECT first-name, last-name
FROM employees
ORDER BY salary DESC;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04. Retrieve the top 3 highest-paid employees

```
SELECT id, first_name, last_name, salary
FROM Employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

05. Find Employees in the IT department

```
SELECT id, first_name, last_name
FROM Employees
WHERE department = 'IT';
```

id	first_name	last_name
1	Alice	Green
5	Eva	Naidoo

06 Find employees in Finance with salary > 60 000

```
SELECT id, first_name, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000;
```

id	first_name	salary
3	Cathy	65000

02 Find employees in HR or Marketing

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	Dave	Mekaena	Marketing

08 Find employees not in IT

```
SELECT id, first-name, last-name, department
FROM Employees
WHERE department != 'IT';
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	Dave	Mekaena	Marketing

09 Find Employees in IT, HR or Finance using IN

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 Find employees in ~~Finance~~ IT with salary > 65000 & city Johannesburg

```
SELECT id, first-name, last-name, department, salary, city  
FROM employees  
WHERE department = 'IT'  
AND salary > 65000  
AND city = 'Johannesburg';
```

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg